CONNECTING CLAUDE TO POWER BI

A Student's Guide to the Power BI Modeling MCP Server

Estimated setup time: 10–15 minutes

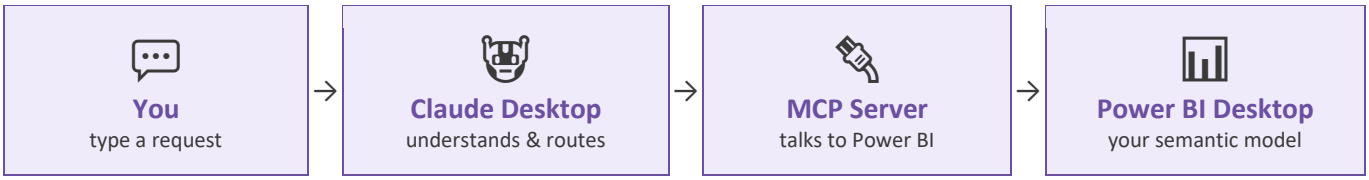


Figure 1 — How the pieces fit together

This guide walks you through connecting Claude Desktop to Power BI Desktop using Microsoft's official MCP (Model Context Protocol) server. Once set up, you can ask Claude in plain English to explore, document, and even build parts of your semantic model, tables, measures, relationships, and DAX, without writing the DAX yourself.

What You'll Need

- ☐ A Windows 10 or 11 computer with admin rights
- ☐ Visual Studio Code (free download)
- ☐ Claude Desktop (free download)
- ☐ About 2GB free disk space and an internet connection

BEFORE YOU CONTINUE

Complete **Setting Up Your Lab** first, including downloading the course files and opening **Sales Report Raw.pbix**. Keep that file open and Power BI Desktop running while you work through this guide.

TIP

The username “Student” and the extension version number in this guide are examples. Replace them with the values shown on your own computer.

The 60-Second Explanation

MCP (Model Context Protocol) is like a universal adapter. It lets Claude “plug into” apps it wouldn’t normally be able to touch, in this case, Power BI. The MCP server runs quietly on your own computer, and Claude sends it

instructions in the background. Your data never leaves your machine except for the specific details you ask Claude about.

1 Install Visual Studio Code

- Go to code.visualstudio.com
- Download the Windows version
- Run the installer with default options
- Once installed, you can leave it closed, you won't need to write any code in it

2 Install the Power BI Modeling MCP Extension

- Open VS Code
- Press Ctrl+Shift+X to open the Extensions panel
- Search for: Power BI Modeling MCP
- Click Install on the extension published by Microsoft

3 Find Your MCP Server Path

- Open File Explorer
- Navigate to: C:\Users\Student\.vscode\extensions\
- Find the folder starting with analysis-services.powerbi-modeling-mcp
- Open it, then open the server subfolder
- Confirm you see a file named powerbi-modeling-mcp.exe
- Right-click powerbi-modeling-mcp.exe → Copy as path, then save the full path for Step 6

Example path (yours will differ slightly):

```
C:\Users\Student\.vscode\extensions\analysis-services.powerbi-modeling-mcp-0.4.0-win32-x64\server\powerbi-modeling-mcp.exe
```

Figure 2 — Save this path somewhere; you'll paste it in Step 6

4 Install Claude Desktop

- Go to claude.ai/download
- Download and install the Windows version
- Open Claude Desktop and log in

5 Open the Claude Configuration File

- In Claude Desktop, click the menu (three lines, bottom-left)
- Go to Settings → Developer tab
- Click “Edit Config”, this opens claude_desktop_config.json in a text editor

FILE LOCATION

%APPDATA%\Claude\claude_desktop_config.json

6 Add the MCP Server Block

Copy the full JSON block below into your config file. Before saving, replace only the bold example path on the "command" line with the full path you copied in Step 3. Keep the quotation marks, commas, braces, and all other text unchanged. If your config file already contains settings, add the mcpServers entry within the existing outer braces rather than creating a second separate JSON block.

```
{
  "mcpServers": {
    "powerbi-modeling-mcp": {
      "command": "C:\\Users\\Student\\.vscode\\extensions\\analysis-services.powerbi-
modeling-mcp-0.4.0-win32-x64\\server\\powerbi-modeling-mcp.exe",
      "args": ["--start"],
      "type": "stdio"
    }
  }
}
```

Figure 3 — The mcpServers block, using the dummy username "Student"

WATCH OUT

Windows file paths use a single backslash (\), but JSON requires every backslash to be doubled (\\). Copy the path from File Explorer, then double each backslash by hand or with find-and-replace before pasting.

COMMON ERROR

A stray extra { or } is the most common cause of a "Could not load app settings" error. If your file already has other content, make sure mcpServers is added as a sibling entry with a comma before it, never as a second, separate { ... } block floating on its own.

Save the file once you're done editing.

7 Restart and Verify

- Fully quit Claude Desktop (check it isn't still running in the system tray)
- Reopen Claude Desktop
- Go to Settings → Developer tab again
- Confirm "powerbi-modeling-mcp" appears with a status of Running or Connected

8 Connect to Power BI

- Confirm "Sales Report Raw.pbix" is still open in Power BI Desktop

- In Claude Desktop, type:

```
Connect to the open Power BI Desktop file.
```

Claude should reply confirming it has connected, you're ready to go!

> **TIP**
> The first time Claude connects, grant permission to use “Connection Operations” from powerbi-modeling-mcp and accept the EULA. Later commands may request other permissions, such as “DAX Query Operations”; grant the requested permission to continue. Start with a small, read-only request such as “List all the tables and measures in this model.” You can then try “Suggest better names for any columns that break naming conventions” before asking Claude to change anything.

🛠 Troubleshooting Cheat Sheet

Problem	Likely Fix
“Could not load app settings”	Your config file has a JSON syntax error, usually a missing comma or a stray extra { or }.
“PowerBI Modeling MCP is not running”	Check every backslash in your path is doubled (\\), then fully restart Claude Desktop.
“File not found”	Your extension's version number differs from the example. Copy the exact folder name from File Explorer instead of typing it.
“Connection refused”	Make sure a Power BI Desktop file is actually open before asking Claude to connect.

🏁 Quick Recap

- Install VS Code → install the extension
- Copy the full powerbi-modeling-mcp.exe path from the extension's server folder
- Install Claude Desktop → open its config file
- Add the mcpServers block with your path, doubling every backslash
- Restart Claude Desktop completely and verify it says Connected
- **Complete Setting Up Your Lab first**, then keep “Sales Report Raw.pbix” open, ask Claude to connect, and grant any requested permissions

You're now ready to explore Power BI with Claude, happy modeling!